Sagar Gupta

ProServe (Cloud Consultant) - DevOps/MLOps | Amazon Web Services (AWS) | Hyderabad, India

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Summary

MLOps Engineer with two years at AWS Professional Services architecting production ML platforms and DevOps automation for enterprise financial-services and technology customers. Graduated the internal MLOps SME Program by shipping a SageMaker Pipelines platform for medical image classification – blue-green endpoints, Model Monitor drift detection, and Terraform + CodePipeline provisioning – with automatic rollback on drift breach. Deep IaC and CI/CD foundation from DevOps engagements at RWS, State Street, and DTCC. Built LeetCode Rating Predictor: TensorFlow DNN on 121K+ contest records, MAE ∼7.8, GraphQL data pipeline, FastAPI serving on Render. Open-source contributor to Feast, Apache Airflow, and Prefect. 5x AWS certified. LeetCode Guardian – 2166 peak, 1300+ problems solved.

Experience

Amazon Web Services (AWS) Aug 2024 – Present
ProServe (Cloud Consultant) - DevOps/MLOps Hyderabad, India

- **MLOps Pipeline – SME Program Graduate (Jun 2025 – Aug 2025):** Architected end-to-end SageMaker Pipelines platform for medical image classification – training, model registry, blue-green canary deployment with automatic rollback on error and latency alarms, auto-scaling, and PSI drift detection. Provisioned full infrastructure as code via Terraform + CodePipeline; exposed inference through API Gateway + Lambda with a CloudFront frontend. Generalized and published by AWS as an official open-source sample (aws-samples, MIT-0): 86 Terraform files across 4 stacks and 13 reusable modules, clinical quality gates, and closed-loop drift retraining.
- **Lead DevOps Consultant – RWS (Feb 2026 – Present):** Built the AWS foundation for a data-centre exit and a second landing zone entirely in Terraform: Organizations OUs with Control Tower baselines, SCPs, tag policies, and least-privilege account baselines; 4,000+ resources across 20+ Terraform Cloud workspaces, none failing. Cut per-account setup by ∼90% (3+ days to under 4 hours). Delivered two Transit Gateway hubs with Network Firewall inspection and VPN/Direct Connect, plus self-service onboarding that gives each new workload its route table within 15 minutes.
- **DevOps Consultant – State Street (Oct 2024 – Aug 2025):** Built Terraform for about 10 AWS services with automated tests (tftest + PyTest) on the regulated landing zone team, part of the path that opened 38+ services to application teams. Designed Amazon Detective enablement via an AFT customization, with organization-wide auto-enable in Terraform. Worked on the security solution behind 1,000+ preventative CCMv4.0 controls (team result). Project closed with a 10/10 CSAT.
- **DevOps Consultant – DTCC (Aug 2025 – Nov 2025):** Modernized legacy Terraform to cloud-native IaC patterns on high-sensitivity financial workloads. Used Amazon Q to accelerate code analysis and migration assessments.
- **Internal Impact:** Published a second AWS sample, Terraform for AWS Organizations governance on Control Tower (aws-samples, MIT-0), covering six policy types, Control Tower controls, and delegated admin for 15 services. Published APG pattern "Secure PR Previews and Approval-Gated Releases for Amplify Gen 2" (plus 2 more peer-reviewed in publication pipeline). 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). Delivered 2 org-wide tech talks on Terraform testing and GitOps pipeline gating. Associate Speaker Certified.

Amazon Web Services (AWS) Mar 2024 – Jul 2024
Associate Professional Services – DevOps (Internship) Hyderabad, India

- Migrated on-premises application to AWS ECS with blue-green deployment via Terraform, CodePipeline, and CodeDeploy – zero-downtime releases. Instrumented X-Ray and CloudWatch distributed tracing for production observability.

Ikarus-3D May 2023 – Jul 2023
Software Developer Associate (Internship) Mohali, India

- Containerized production application on AWS ECS/ECR with GitHub Actions CI/CD; deployed SonarQube on EC2 + RDS to embed continuous code quality gates into the delivery pipeline.

Industry Certifications (5)

AWS: Solutions Architect – Associate | Developer – Associate | Machine Learning Engineer – Associate
AWS: AI Practitioner | Cloud Practitioner

Education

National Institute of Technology Warangal, Telangana Sep 2021 – Jul 2024
Master of Computer Applications (MCA) | NIMCET AIR 208 CGPA: 8.38

Sanskar College of Professional Studies (DAVV University), Indore May 2018 – Aug 2021
Bachelor of Computer Applications (BCA) CGPA: 7.34

Projects

LeetCode Rating Predictor Dec 2023
Python, TensorFlow, FastAPI, React, GraphQL, Docker, Render GitHub | Live
- ◇ Trained dense neural network on **121K+ contest records** for post-contest rating prediction with **MAE ∼7.8** on held-out set. Automated feature engineering pipeline over LeetCode GraphQL API.
- ◇ Productionized full MLOps lifecycle: containerized FastAPI inference service, React frontend, backend and frontend test suites, Docker + Render deployment with health checks.

Bedrock Multi-Model MCP Apr 2026
TypeScript, MCP Protocol, AWS Bedrock Converse API, Nova Canvas, Nova Reel GitHub
- ◇ MCP server routing 500+ Bedrock foundation models across text (Llama 4, Mistral, Nova, Claude, Cohere, DeepSeek R1), image (Nova Canvas, SD 3.5, Titan), and video (Nova Reel async S3) – with `bedrock_compare` side-by-side inference evaluation across 2–5 models.

Blue-Green AWS Terraform Jul 2025
Terraform, AWS ECS on EC2, ALB, CodePipeline, HCL GitHub
- ◇ Production-grade IaC for zero-downtime blue-green deployments on ECS (EC2 Auto Scaling Group) with 15 Terraform modules – directly applicable to ML model serving with progressive traffic shifting and automated rollback.

GroundZeroNN 2024
Python, NumPy GitHub
- ◇ Feedforward neural network built from scratch in NumPy – forward/backward propagation, activation functions, mini-batch gradient descent. **79.8% test accuracy on MNIST**, no ML frameworks.

Ledger Sync Jan 2026
TypeScript, React, FastAPI, SQLite, AWS Bedrock, OpenAI, Anthropic GitHub | Live
- ◇ Data-intensive personal finance app with SHA-256 deduplication pipeline, anomaly & recurring transaction detection, and multi-provider LLM chatbot (15 read-only tool calls) over Excel/CSV imports.

Other Notable Projects: StockSage (LSTM stock forecasting), SelfHub (MCP personal memory server), Claude Cost Optimizer (30–60% AI cost reduction plugin), GitScope (Chrome extension), Instagram Autopilot (Bedrock Nova Canvas/Reel).

Open Source Contributions

- **axios** (109K☆) – Synthesize AxiosError message from AggregateError on dual-stack connection failures (#11059, merged)
- **Apache Airflow** (44K☆) – Add `template_fields` to SalesforceBulkOperator for Jinja templating (#63109, merged)
- **PrefectHQ/prefect** (21K☆) – Document custom deployment steps as Python functions (#20956, merged)
- **aws-ia/terraform-aws-vpc** – Root-caused the Cloud WAN attachment-replacement bug; fix co-authored into v4.8.0 (merged)
- **hashicorp/terraform-provider-aws** (10K☆) – IoT `cloudwatch_metric` substitution template validation (#46867, open)
- **feast-dev/feast** (9K☆) – Claude Code agent skills for Feast feature store (#6081, merged)
- **awslabs/agent-plugins** – defusedxml extension guard (#132, merged)
- **awslabs/mcp** – Fix Kendra documentation menu to match the source directory (#2607, merged)
- **expressjs/multer** – Stop files skipped by `fileFilter` from consuming `maxCount` (#1426, merged)

Technical Skills

MLOps & ML Platform: SageMaker Pipelines, Model Registry, Endpoints (real-time & async), Model Monitor, Drift Detection, Blue-Green Deployment, A/B Traffic Shifting, Feature Stores (Feast)
ML Frameworks: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Experiment Tracking, Hyperparameter Tuning
CI/CD for ML: CodePipeline, CodeBuild, GitHub Actions, Docker, Model Versioning, Automated Retraining Triggers
Cloud & Infrastructure: AWS (Lambda, API Gateway, ECS Fargate, CloudFront, EventBridge, S3, CloudWatch), Terraform, HCL, tftest
Data & Serving: FastAPI, GraphQL, REST, Data Pipelines, SHA-256 Deduplication, SQLite, X-Ray, CloudWatch Metrics
Languages: Python, HCL, SQL, TypeScript, JavaScript, Bash

Achievements

- **MLOps SME Program Graduate** (2025) – shipped SageMaker Pipelines platform with blue-green endpoints and drift detection.
- Published APG pattern "Secure PR Previews and Approval-Gated Releases for Amplify Gen 2" on AWS Prescriptive Guidance (plus 2 more peer-reviewed in publication pipeline).
- 5x TFC Ambassador (AI/ML, NGDE Graduated, CloudOps, Agentic AI, Serverless). Associate Speaker Certified.
- LeetCode Guardian – 2166 peak rating, 1300+ problems solved, 107 contests. GeeksforGeeks: 500+ DSA problems. HackerRank: 6-Star Problem Solving, 5-Star C++.
- Rank 1289 (Top 5%) – Google Kick Start Round E '22. 1st place: Coding Cauldron '23 (NIT Trichy), QFiesta IGNOSI '22 (NIT Warangal).
- NIMCET AIR 208 – MCA at NIT Warangal, First Division with Distinction.